Nicolas Duran Garces

Backend Engineer | AI Engineer | Python, Java, TypeScript, AWS
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Summary

Backend engineer with 5+ years building production systems in Python (FastAPI), Java (Spring Boot) and TypeScript (NestJS) on Amazon Web Services (AWS). Founder of Bheex, an AI customer-service SaaS in production with LLM agents, retrieval-augmented generation (RAG) over a vector database, WhatsApp and voice AI channels. Experienced in microservices, event streaming, large-scale browser automation and IoT ingestion, and a former Software Engineering professor. Seeking a senior backend or AI engineering role with ownership of architecture.

Core Competencies

- **Backend:** Python (FastAPI, Django, Flask), Java (Spring Boot 3, microservices, hexagonal architecture), TypeScript/Node.js (NestJS, Express), REST APIs.
- **AI and LLMs:** LLM integration (OpenAI API, Gemini), agents with tool/function calling, RAG with embeddings and Qdrant, voice AI (Twilio, ElevenLabs text-to-speech, speech-to-text), Model Context Protocol (MCP), Claude Code, N8N.
- **Cloud and DevOps:** AWS (Lambda, EC2, ECS Fargate, IoT Core, Amplify, CloudFormation, CloudFront), Azure, Docker, Terraform, Nginx, CI/CD, Serverless Framework.
- **Data and Messaging:** PostgreSQL, MySQL, Redis, Prisma, Kafka (AWS MSK), RabbitMQ, BullMQ, MQTT (Mosquitto).
- **Automation:** Selenium Grid on Docker, web scraping, robotic process automation (RPA), Chrome extensions.
- **Frontend:** React, Next.js, Vite, Tailwind, React Native, Flutter Web.

Professional Experience

Founder and AI Engineer | Bheex (own company) | Remote | 2026 - Present

- Built and operate a multi-tenant AI customer-service SaaS in production on AWS EC2, with NestJS 11, Prisma, Next.js 15, BullMQ workers, PostgreSQL 16, Redis and Docker CI/CD.
- Designed the LLM agent pipeline on Gemini: intent router plus multi-turn agent with tool/function calling and flow routing configured from the admin UI.
- Implemented RAG end to end: document ingestion (PDF, Word, Excel, CSV), chunking, embeddings and a Qdrant vector database filtered per tenant, with cited answers.
- Shipped WhatsApp (Twilio webhooks) and voice AI calls with speech-to-text, ElevenLabs text-to-speech and live AI-to-human handoff without dropping the call.
- Added guardrails: read-only SQL tools (SELECT-only, table whitelist, forced LIMIT, timeouts), SSRF protection, AES-256-GCM encrypted secrets and tenant isolation.

Backend Developer | Software company | Remote, Colombia | Jul 2026 - Present

- Build the backend of a lottery and gaming platform: about 17 Spring Boot 3 / Java 17 microservices in hexagonal architecture plus a Flutter Web front end, currently in user acceptance testing (UAT).

Full Stack / Backend Engineer (Freelance) | Independent | Remote | May 2026 - Present

- U360 (ed-tech): NestJS 11 and React 19 monorepo on AWS (CloudFormation, ECS, CloudFront); completed a full data-model migration and moved the student portfolio from mock local storage to a real backend.
- FastAPI services consuming Kafka (AWS MSK) event streams for a payments network and normalizing messages into internal contracts.

Software Engineer | BetterWay Devs | Remote | May 2024 - Present

- Develop a price-match and refund bot: Selenium on AWS ECS Fargate driven by a FastAPI control API, Terraform and GitLab CI/CD, with a Python LLM backend calling the OpenAI API.
- Built an operations API and dashboard tracking hourly concurrency, queue time, success and cancellation rates.
- Evolve Python and AWS Lambda automation platforms and maintain the cloud infrastructure for large-scale retail scraping and data processing.
- Modernize legacy code and orchestrate services for high availability; accelerate delivery with Claude Code, MCP and LLM tooling.

Full Stack Developer Lead, Osmed Gateway | Grupo de Innovacion EAM | Armenia | Sep 2024 - Sep 2025

- Led an end-to-end HealthTech platform for clinical records and scheduling with React, Node.js and PostgreSQL, deployed on AWS.
- Built the native iOS app (React Native) and implemented role-based access control and real-time medical scheduling.

Cloud and IoT Solutions Architect, EDEQ | Grupo de Innovacion EAM | Armenia | Jun 2024 - Sep 2025

- Designed the serverless architecture (AWS Lambda and Docker) for the Quindio Inteligente project.
- Built a Python backend for high-volume IoT ingestion over MQTT (Mosquitto) for real-time monitoring, and reduced cloud cost with ephemeral containers and on-demand services.

Software Engineering Professor | Institucion Universitaria EAM | Armenia | Aug 2023 - Jun 2025

- Taught Enterprise Applications (Java Spring Boot, Docker), a Microservices elective (Python FastAPI, AWS Serverless/Amplify), Programming Logic and Software Design; mentored students on clean architecture and cloud deployment.

Software Developer | Atlanticsoft | Colombia | Aug 2021 - May 2024

- Built the critical backend of an automated warehouse sorting conveyor (WMS), bringing response times under 1 second to keep physical parcels in sync.
- Migrated Chrome-extension automations to Dockerized Selenium clusters on Azure, sharply reducing order cancellation rates.
- Delivered warehouse control interfaces in Bubble (no-code) integrated with the backend.

Software Developer (Internship) | Atlanticsoft | Armenia, Colombia | Feb 2021 - Aug 2021

- Built Chrome extensions automating bulk purchasing on Amazon, BestBuy and Walmart, and AWS Lambda functions to store transaction and purchase logs.

Projects

- **Atlas, distributed e-commerce (2025):** Spring Boot microservices with RabbitMQ and Eureka, SAGA transactions, Resilience4j circuit breakers, rate-limited API Gateway, Prometheus, Grafana and Zipkin.
- **Caturro Cafe ecosystem (2025):** four containerized systems (POS, loyalty, payroll scheduler, landing) with Next.js, FastAPI, Express, Prisma/MySQL, Docker and Nginx.

Education

Software Engineering | Institucion Universitaria EAM, Armenia | 2018 - 2022
Software Development Technologist | Institucion Universitaria EAM, Armenia | 2018 - 2021
Software Programming Technician | SENA, Colombia | 2016 - 2017

Certifications

IBM AI Developer Specialization (IBM) | AWS Academy Graduate - Cloud Foundations (AWS) | Platzi: Cloud Digital Leader, Serverless Framework on AWS, FastAPI, Docker, High-Concurrency Architecture, Software Architecture, Web Scraping with Python, N8N

Languages

Spanish (native) | English (B1: fluent technical reading and writing)